Advanced Microeconomics

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Problem Set 7: Applications of decisions under uncertainty

1 Application

1) A person with $U(w) = w^2$, where w is the person's income, is facing the following lottery: there are four tickets sold, of which only one wins the prize of \$100.

a) Calculate the expected value (E(w)) of the lottery.

b) Give the definition of and calculate the certainty equivalent (C) of this lottery.

c) Illustrate your solutions to a) and b) graphically. Also use your graphical representation to explain the term risk premium.

d) Compare and discuss results in parts (a) and (b). What are this person's preferences toward risk?

2) A's utility function is $U = \sqrt{w}$, where w is her wealth. She owns a bakery that will be worth 70 or 0 dollars next year with equal probability.

a) Suppose her firm is the only asset she has. What is the lowest price P at which she will agree to sell her bakery?

b) Redo part (a) assuming that she has \$100 safely stored under her mattress.

3) An individual plays soccer in a team in Brazil. If he does not get injured by the end of the next season, he will get a professional contract in Europe, worth 10,000 EUR. If he gets injured, he gets a contract as a firefighter in São Paulo, worth only 100 euros. The probability of getting injured is 20%. With 80% probability he will get through the season unscathed.

a) What is his expected utility if his utility function is $u(x) = \ln(x)$?

b) What is the certainty equivalent for this individual to the lottery above? What is the risk premium?